New Techniques for Analyzing Block Cipher Differentials

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Abstract

Differential cryptanalysis is one of the most powerful attacks on block ciphers. However, estimating the probability of a differential (i.e., the probability that a given input difference leads to a given output difference) remains a challenging problem for modern block ciphers.

In this talk, I will describe new techniques for analyzing differentials under the standard assumption of independent round keys. These techniques are developed using tools from the theory of Markov chains. As an application, I prove that every nonzero differential in 8-round AES has probability very close to the ideal value for a pairwise independent permutation.